Installing the OWEAdesign R package and running its web app on your computer

OWEAdesign (formerly owea) finds optimal experimental designs (D- and A-optimal, linear and generalized linear models, exact and compound designs). It is written partly in C++, so installing it requires a compiler once. Follow the four steps in order; the whole process takes about 10 minutes with an internet connection. Package page: https://github.com/minyangstat-eng/OWEAdesign.

Step 1. R (version 4.0 or newer) and RStudio

- Install R from https://cran.r-project.org/ and, optionally, RStudio Desktop from https://posit.co/download/rstudio-desktop/.
- Check: in the R console, `R.version.string` should report 4.0 or higher.

Step 2. A C++ compiler (once per computer)

- **Windows:** install Rtools from https://cran.r-project.org/bin/windows/Rtools/, choosing the version that matches your R (Rtools44 for R 4.4.x, Rtools45 for R 4.5.x). Keep the default options and restart RStudio afterwards.
- **macOS:** open Terminal, run `xcode-select --install` and accept the dialog (installs Apple's command-line tools, 1–2 GB).
- **Linux (Ubuntu/Debian):** `sudo apt-get install build-essential r-base-dev` in a terminal.
- **Check:** in R run `install.packages("pkgbuild"); pkgbuild::has_build_tools(debug = TRUE)`. It must print TRUE before you continue.

Step 3. Install OWEAdesign and its dependencies

In the R console (compilation takes 1–3 minutes; long output and warnings are normal, errors are not):

```
install.packages(c("remotes", "Rcpp", "RcppArmadillo", "shiny", "DT", "lpSolve"))
remotes::install_github("minyangstat-eng/OWEAdesign")
```

If asked *These packages have more recent versions available. Which would you like to update?*, type 3 (None). Then confirm the installation:

```
library(OWEAdesign)
packageVersion("OWEAdesign")  # 0.6.0 or higher
```

Step 4. Run the web app

```
library(OWEAdesign)
run_owea_app()
```

A browser window opens with the point-and-click app. The R console is busy while the app runs; press **Esc** in the console (or RStudio's red stop button) to stop it. If no window opens, paste the address printed in the console (`http://127.0.0.1:XXXX`) into your browser.

If something goes wrong

- `compilation failed` / `make: command not found` (Windows): Rtools is missing or does not match your R version — reinstall it, restart RStudio, repeat the Step 2 check.
- `xcrun: error` / `clang: command not found` (macOS): command-line tools missing — repeat Step 2.
- `package 'RcppArmadillo' is not available`: run `install.packages("RcppArmadillo")` on its own, then repeat Step 3.
- `run_owea_app()` reports a missing package: `install.packages(c("shiny", "DT", "lpSolve"))`.
- Everything else: open an issue at https://github.com/minyangstat-eng/OWEAdesign/issues with the error message, your operating system and `R.version.string`.